A Minimal Epistemic Core

Differentiation, Receptive Dependence, and Registration

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ABSTRACT. This paper states the minimal structure of the basic form of any epistemic relation: the relation in which something is perceived, or taken up in any way, prior to concepts and descriptions, whatever the domain in which it is realized. It rests on two principles. The first is that to be some way in particular is to differ. The second is that the relation is receptive: that by which something is taken up is due to what is taken up. From these it follows that what such a relation is of is a difference; that the registering is itself a difference, which holds at least partly in virtue of the first; and that both are differences between actual terms, since only what holds can be that in virtue of which something holds. Alternatives, and any repertoire of possibilities, are argued to be products of registration and not preconditions of it. Registration is then defined in a strictly structural sense that presupposes neither understanding, nor representation, nor order, nor a pairing of terms, nor a knowing subject. The account uses two notions without analysing them, a field's bearing a contrast and one contrast's holding in virtue of another, and binds each by stated conditions. The conditions are formal: they presuppose no spatial, temporal, or physical structure. They are necessary, not sufficient, with the necessity of an analysis: they hold of whatever is a receptive relation to something in particular. The core is a schema, which states what any dependence must satisfy to count as registration and leaves to each domain what the dependence consists in; two realizations, one causal and one not, show that it is met. They also show that the base of the epistemic is not itself epistemic: the structure is found wherever one difference is due to another, and what makes a registration epistemic lies in what is built on it. The paper lists everything it assumes, by kind, the points at which it can be refused, and what the core does not provide.

1. Introduction

Theories of knowledge ordinarily begin with a subject who believes, judges, or is justified, and ask under what conditions such states amount to knowledge. This paper addresses a prior question: what must be admitted for any epistemic

relation to be possible at all—for anything to be perceived, or taken up in any way. Its answer is a set of necessary conditions, not a sufficient account. It identifies what any theory of perception, belief, or knowledge presupposes, and leaves to such theories what makes a relation epistemic in the full sense. The assessment of judgement, in particular, lies outside its scope.

The guiding premise is that, in its basic form, prior to concepts and descriptions, an epistemic relation is always a relation to something in particular. What such a relation is of is a difference, and the registering is itself a difference that holds in virtue of the first. Section 2 states the premise, derives it from two principles, and says what kind of necessity it has and at which points it can be refused.

The conditions are formal. They are stated without reference to space, time, matter, or causation, and they are meant to hold whatever the domain in which registration is realized, whether perceptual, physical, or linguistic. Examples drawn from particular domains show how the conditions are met there; they do not fix what the conditions are. For the same reason, necessity is claimed only where it can be argued for, and features of the presentation that are conventions rather than commitments are marked as such.

The reconstruction proceeds by minimal commitment. Each element is introduced only at the point where the account cannot proceed without it, and each is accompanied by a statement of what it does not establish. One methodological caution applies throughout. The exposition necessarily employs ordinary logical resources—identity, distinction, relation, and inference—but using them to state the commitments does not attribute them to what the commitments describe, and in particular not to a knowing subject. A distinction drawn in the exposition is not thereby a distinction drawn by its object. In the same way, the exposition uses modal words to say what the conditions leave open; this attributes no possibilities to what it describes.

The proposal has close neighbours. Dretske (1981) analyses the information carried by a signal in terms of nomic dependence and conditional probability, and separates it from meaning and belief. Floridi's diaphoric definition of data takes a lack of uniformity—a difference—as the most elementary constituent of information (Floridi 2011), and his informational structural realism holds that the relation of difference is constitutive of its relata (Floridi 2008). Smith's metaphysics of presence makes *registration* its central notion, as the way in which a world comes to be taken as one of objects (Smith 1996).[1] Enactivist accounts describe cognition as the structural coupling of a system with a medium, both of which belong to a larger organization (Maturana and Varela 1987; Thompson 2007). The present account differs from each in three respects. It separates the contrast that is registered, the contrast that registers it, and the dependence

1. The present use of the term is narrower. Smith's registration is performed by subjects and yields objects; registration here presupposes neither and concerns only the dependence of one contrast on another.

that connects them, and it says of each element whether it is asserted as a principle, argued, stipulated, or left unanalysed. It states that dependence between contrasts of actual terms, without probabilities, causes, or alternatives on either side, so that it applies beyond the physical domain and requires no repertoire of possibilities. And it distinguishes the direction of dependence from any order of occurrences, leaving open what kind of order further epistemic operations require.

Section 2 states the guiding premise and the two principles from which it follows. Section 3 introduces the notion of a field's bearing a contrast and the condition of method under which the reconstruction proceeds. Section 4 defines articulations and their realization, and states two consequences. Section 5 states receptive dependence exactly, with the roles it defines; Section 6 draws from it the registering contrast and its bearer. Section 7 defines registration on that basis and states what it does not include. Section 8 works the schema in two cases, and Section 9 shows that the resulting structure assumes no order. Section 10 lists everything the account assumes, what it rules out and what it does not include, its limitations, and the questions it leaves open.

2. The Guiding Premise

The account starts from a premise in two clauses.

> (P) GUIDING PREMISE
>
> *In its basic form, an epistemic relation is a receptive relation to something in particular. (P1) What it is of is a difference: X as it differs, in some respect, from another actual term. (P2) That by which it is of that difference, the registering, is itself a difference, and it holds at least partly in virtue of the first.*

"Basic" means prior to concepts and descriptions: the relation in which something is perceived or taken up, as distinct from one in which it is described or judged. The word *registration* is reserved for the structure defined in Section 7. The premise says that every basic epistemic relation has that structure; the converse is not claimed, and Section 8 shows that it is false.

The premise follows from two principles. Each is stated, with the reason for holding it and the point at which it can be refused.

Determinacy. To be some way in particular is to differ: for there to be a way that the thing is not. A way that excluded nothing would mark nothing, and a thing that was no way in particular could not be what anything is of. The account does not derive this principle from anything more basic. It is what the account takes "in particular" to mean, and it is applied without exception:

to what a relation is of, to the registering, and to the conditions of whatever receives. It can be refused by holding that a thing may be some way in particular while differing from nothing, actual or not, in that respect.

Receptivity. The relation under study is the one in which something is taken up, and what is taken up is that to which the taking-up is due. A registering holds, at least partly, in virtue of what it registers. This principle marks out which relation is being analysed. Its content is shown by what falls short of it. Two contrasts may be alike in any respect; their positions may correspond one to one; one may stand before or above the other. Each of these relations holds between the two by how each of them is. If likeness or correspondence sufficed, every contrast would be of every contrast like it, itself included; if an ordering sufficed, of everything it stands before. In neither case would anything be received by anything. Agreement by arrangement falls short in a different way. Where two contrasts answer to one another because both are due to a third, or because something has fitted the one to the other, whatever aboutness there is in the case is owed to a dependence somewhere in it: of each contrast on the third, or of the fitting on the contrast that is fitted to. Where there is no dependence anywhere, the agreement is coincidence, and a coincidence is of nothing in particular. The principle can be refused by holding that an epistemic relation, in its basic form, need not be receptive. The account regards that as a change of subject rather than a rival answer.

From the principles to the premise. By determinacy, what a relation is of is some way in particular, and so differs in some respect. By receptivity, the registering is due to what it is of; and what something is due to must itself hold. So what is taken up is a difference that holds, which is a difference between terms that are both actual. This is the first clause. It is of the difference, X as it differs, and not of X apart from its differing, because by determinacy X's being some way in a respect is nothing apart from its position among the differences in that respect. The registering, in turn, is some way in particular, and it holds; so by the same reasoning it is a position in a contrast between actual terms; and since a position is nothing apart from its contrast, it is that contrast which is due to the first. This is the second clause.

Why actual terms only. That every contrast in the account is between actual terms is thus a consequence, not a separate choice. It is worth stating directly, because alternatives, ways that nothing actual is, are often taken to be where an epistemic relation begins. There are three places where alternatives might enter, and none of them is prior to registration.

They might enter in what is taken up: the thing is this way rather than another way that nothing actual is. But an alternative cannot be that in virtue of which anything holds. Where a term differs from nothing actual in a respect, there is in that respect no difference that holds, and so none to be received.

They might enter in the dependence: K' depends on K when K' would not have held had K not held. But dependence and what would have held otherwise

come apart in both directions. K' may be due to K although, had K not held, something else would have brought K' about; and neither of two contrasts may hold without the other although neither is due to the other, as when both are due to a third. What would have held otherwise is evidence of dependence, and often the best evidence there is. It is not what dependence consists in.

They might enter in the receiver, as a repertoire against which the actual case is singled out. But a repertoire has to be held, and what holds it is a receiver with actual conditions that stand for what is not the case. Such a receiver already registers and retains. A repertoire is therefore a product of registration, not a precondition of it.

Three things, accordingly, are not basic and are not provided by the core: a relation to which way something is as against an alternative; a relation to anything of a term that differs from nothing actual in the respect concerned; and a relation to the fact that two terms do not differ. The claim is not that these do not occur. It is that each requires what is built on registration, a repertoire or a pairing of terms with conditions, and the full defence of that claim belongs to the stages that build them. The word "actual" is used without analysis, for what is the case as opposed to an alternative that is not. It is not tensed: nothing is said about when.

The standing of the premise. The premise is an analysis of what it is for a relation to be a taking-up of something in particular. Its necessity is that of an analysis: it holds of whatever the notion applies to, of whatever kind, and nothing in it depends on what sort of thing receives. It is not claimed that any such relation occurs, nor that the conditions hold of anything apart from one. It may be objected that reasoning of this form shows only how we must think of epistemic relations, not how they are (Stroud 1968). That objection applies to arguments that pass from features of our thinking to features of the world. The premise makes no such passage. It says what a relation must be in order to be a taking-up of something, and it leaves open whether, where, and by what anything is taken up. What remains true in the objection is that an analysis is answerable to the notion analysed, and that we know the notion from our own case; that is why the two points of refusal are stated.

The premise concerns the basic form only. A claim may be about something by description, about a future event, or about a mathematical object, and the premise does not cover these cases; whether they are built from the basic form is left open (Section 10). "In particular" does not mean "one only". It will turn out that, in the sense to be defined, a contrast registers every contrast it holds partly in virtue of (Section 5); particularity requires that there be a fact about what is registered, not that it be single.

3. Fields and Contrasts

The account uses one notion without analysing it: that something *bears a contrast* in some respect. It is left unanalysed because any definition of it would use the notion of a difference, or one no clearer. A notion so used must still be bound by stated conditions, so that a misuse of it can be shown. There are four. A contrast is always in some respect; the account does not say what respects there are, nor whether bare distinctness is one. It has two positions, and no position differs from itself. Its positions are both actual. And it does not divide what bears it exhaustively: the positions of a contrast need not be all there is. A contrast is always borne by something. Whatever bears a contrast is, insofar as it does, a *field*. To be a field is thus an aspect, not a kind of thing: the term implies neither a spatial region, nor a collection of discrete objects, nor something already presented to a subject. Let F designate the field under consideration. The designation fixes the scope of the inquiry. It does not establish a boundary intrinsic to reality: F identifies what the account concerns, and any claim about its actual boundaries would require separate justification.

That F bears a contrast is not a modal claim. It does not say that F can be differentiated, or that it admits differentiations which it does not have; it says that there is a difference. Nothing is thereby said about which contrasts F bears or of what form, about whether some differentiation of F is uniquely privileged, or about whether what differs would be objects, qualities, regions, values, or something else. Nor is anything implied about change or about any order. Possibilities and capacities of a field are not thereby denied. They are not part of the core, and if a later stage of the theory needs them, they must enter as commitments of their own, stated as such.

The reconstruction is subject to one condition. It is a condition of method, not a claim of the core.

SCOPE OF THE RECONSTRUCTION

The reconstruction articulates only fields taken to bear the contrasts it specifies.

The reason is that the reconstruction concerns a differentiation attributable to its target, rather than one supplied entirely by an unrelated description; a contrast is attributable to a field when the field bears it. This is a necessary condition for the route of explanation pursued here. It is not arbitrary: by receptivity, what a registering is due to must hold, so a difference that held only in a description could be registered only as a difference in the description. The account does not argue against the view that all differentiation is supplied by descriptions. It states what that view would cost: if it held, no contrast would be attributable to a target, and the reconstruction would have nothing to articulate.

No knowing subject is involved at this point either: that F bears a contrast does not imply that anything differentiates it.

4. Articulation

An account speaks of a particular differentiation of F by means of an *articulation*, written

$$\langle a, \delta, b \rangle_F,$$

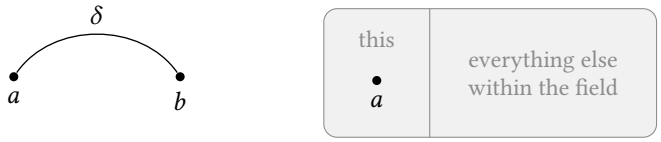
where a and b are its *terms* and δ is its *respect*. Its *contrast* is the difference of a and b in that respect; two articulations may share a respect and have different contrasts. Two levels must be kept apart. A *position* is a place in a contrast that a field bears. A *specified term* is what a description picks out, and it may pick out nothing. An articulation is *specified* when its terms and respect are fixed, and it is the account, not F, that specifies it. It is *realized* when its specified terms pick out positions of a contrast that F bears in the respect δ. Without the two levels, every articulation would be realized by definition. By stipulation, two contrasts are the same when they have the same positions and the same respect; no other individuation is assumed.

The terms of a realized articulation are both actual. They are not alternatives of which one is there rather than the other, and the contrast is not a choice between them; it is the difference between two terms that are both actual. The schema has two terms because a contrast is between two positions; a differentiation among more terms is a number of contrasts. Nor does the schema presuppose that a and b exist prior to the contrast as independent objects; it marks positions within it.[2]

Two misreadings of the schema should be excluded. First, δ is not a third item situated between a and b. To treat it as a separator would presuppose spatial organization, whereas δ expresses the differentiation itself. Second, b need not comprise everything that remains once a has been selected. That reading would require an exhaustive partition of F, and a contrast between two terms is not equivalent to a division into "this" and "everything else within the field": the latter adds the requirement that every case within the scope fall on one side or the other. Figure 1 sets the two constructions side by side.

Specification chooses the respect and selects a contrast borne in it. That it does not create the contrast is the condition stated in Section 3. Whether F realizes the specified articulation, and whether anything depends on it, are not

2. The refusal to ground a difference in terms identified before it has a precedent in Deleuze's critique of the subordination of difference to identity (Deleuze 1994, ch. 1). Deleuze also rejects the possible as a mere image of the real; the present account shares the refusal of a stock of possibilities.



(a) contrast (b) exhaustive partition

FIGURE 1. Two constructions over F. (a) An articulation $\langle a, \delta, b\rangle_F$: a and b mark positions, both actual, and δ, drawn as a relation rather than as an item between them, is the respect in which they differ. (b) An exhaustive partition, which adds the requirement that every case fall on one side or the other. Only (b) is drawn as a region, because only (b) carries the extensional commitment that a region suggests; in (a) the placement of the marks has no spatial meaning.

settled by the specification, which can therefore fail. Specification fixes what an inquiry concerns, as F fixes its scope; it does not supply what the inquiry finds.

The condition of Section 3 accordingly leaves out of the reconstruction any respect whose holding depends, even partly, on the specification or its notation: on the order, the labels, or the grouping of the terms. Suppose that F bears a contrast in the respect δ, and let δ^* be the respect: δ, and a is written first. F does not bear a contrast in δ^*, because part of what makes δ^* hold is not in F. A differentiation of this mixed kind is not excluded wholesale. The contrast that F bears, in δ, is within the scope of the reconstruction; the remainder is not.

Two consequences follow at once. The first, recorded as (D), is that every field in which an articulation is realized bears that articulation's contrast, and so in particular every field in which a registered contrast is picked out. This is not a commitment. It follows from the definition of a realized articulation, and it holds of any realized articulation, registered or not. It establishes neither which contrasts a field bears, nor that any field bears a contrast apart from registration, nor any modal claim about F. The second concerns a lone term. A contrast has two positions, both actual. What differs from no actual other term in a given respect is a position in no contrast in that respect, and the core provides for registering nothing of it in that respect.

These qualifications mark the first limit of the reconstruction. That a field bears contrasts does not fix which articulation an account will specify. An account of any particular case must therefore specify the articulation it concerns, and it may fail to pick out anything. Moreover, a realized articulation remains an articulation of F, and nothing established so far implies that it is registered anywhere.

5. Receptive Dependence

That F bears a contrast, and that some other contrast is borne somewhere, leaves the two unconnected. The second clause of the premise requires more: that the one hold in virtue of the other. Let C designate *receptive dependence*. The thesis below states the second clause exactly; it adds no commitment to it.

> (C) RECEPTIVE DEPENDENCE
>
> *A contrast K', borne by a field, holds at least partly in virtue of a contrast K borne by F.*

(C) relates two particular contrasts. An account states an instance of it by specifying an articulation whose contrast is K; the specification belongs to the account, and the dependence does not wait on it.

The phrase "in virtue of" is the second notion that the account does not analyse. The analyses on offer either state it over alternatives to the actual case, as counterfactual analyses do, and so mistake its evidence for its content (Section 2), or belong to one domain, as causal analyses do. Like the first notion, it is bound by stated conditions, given in this section: as it figures in (C), it relates contrasts that both hold; nothing holds in virtue of itself; it has a direction; it may be partial; and whether it holds is not fixed by any description of what it relates. Where the exposition says more loosely that something is due to, or because of, something that is not a contrast, the word is the ordinary one and carries none of these conditions. K' holds because of K, and not merely alongside it. The phrase is not meant to identify C with metaphysical ground; it borrows only two features that the literature on ground has made familiar, namely that dependence of this kind has a direction and that it relates what actually holds, without appeal to what would hold otherwise (Fine 2012). (C) is accordingly not a counterfactual claim. It does not say what K' would be if F were otherwise; it relates two contrasts that both hold.[3]

Dependence must be distinguished from two relations that supply none. Two contrasts may resemble one another without either registering the other,[4] and the terms of one may be paired with the terms of the other, a with r and b with r', without either contrast holding because of the other (Figure 2). Such a pairing can be read in either direction. (C) requires no pairing of terms. It relates one contrast to another, and which position of K' answers to which position of

3. Counterfactual analyses of dependence require alternatives to the actual case. Where contrasts hold necessarily, they meet a further difficulty: on the standard semantics, counterfactuals with impossible antecedents are all vacuously true (Lewis 1973); for an alternative semantics, see Nolan (1997). Neither difficulty affects (C).

4. Goodman (1976, ch. 1) makes the parallel point about representation: resemblance is reflexive and symmetric, whereas representation is neither, so resemblance cannot be what representation consists in.

K is not part of it. Where a pairing is needed, as it may be for registering which way F is (Section 7), it must enter as a further commitment, stated as such.

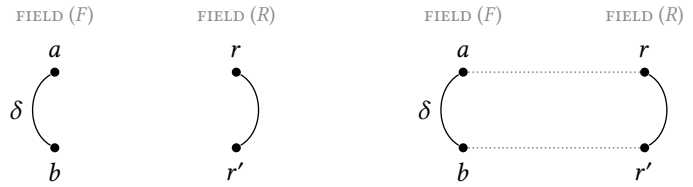
The dependence has a direction. That K' holds in virtue of K is one thing, and that K holds in virtue of K' is another, and the first does not bring the second with it. Nothing holds in virtue of itself; this is stipulated, so that no contrast counts as registering itself. Whether two distinct contrasts can each hold partly in virtue of the other is left open. The premise does not exclude it: it requires that the registering depend on the difference, not that the difference be independent of the registering. If it occurs, there are two instances of C, one in each direction, and each contrast registers the other. A view on which a difference in F is partly made in being registered is therefore not excluded by the core; it is described as registration in both directions. The dependence holds between contrasts, not between fields: two fields may each depend on the other through different contrasts, and K' may hold partly in virtue of K while a further contrast K'' of F holds partly in virtue of K'. Whether the dependence is transitive is likewise left open. If it is not, loops through a third contrast are not excluded; if it is, and no two contrasts depend on each other, the dependence is a strict partial order among contrasts.

The dependence may be partial. K' holds at least partly in virtue of K, and whatever else it is due to may lie in the field that bears K'.[5] The point is captured in Bateson's definition of information as "a difference which makes a difference" (Bateson 1972), anticipated in MacKay's account of information as a distinction that makes a difference (MacKay 1969), which compresses into one phrase what the present account keeps apart: the first difference is K, the second is K', and the making is the dependence C. For the same reason, the content of (C) should not be concealed behind appeals to "systematic" or "relevant" coupling, which import what the argument has not supplied, namely a standard of relevance or a regularity across cases.

Whether (C) holds in an instance is fixed neither by descriptions of the two contrasts nor by what would hold otherwise. Two instances alike in their contrasts, and in any correspondence between their terms, may therefore differ in whether (C) holds: one is registration, the other mere correspondence. Whether (C) in an instance can depend on other actual instances, as in a regularity across cases, is left open.

(C) does not single out one registered contrast. If K' holds partly in virtue of K and partly in virtue of another contrast K^*, it stands in C to both; and if the dependence is transitive, it stands in C to whatever K in turn depends on. At this level a contrast registers every contrast it holds at least partly in virtue of. To narrow this to one is not a condition of registration but a further achievement, for which a later stage of the theory must supply the means.

5. This accommodates the enactivist point that a perturbation triggers, but does not specify, the changes it occasions in a system, which are fixed by the system's own structure (Maturana and Varela 1987).



(a) resemblance (b) correspondence

FIGURE 2. Two relations that fall short of receptive dependence. (a) Resemblance: two contrasts are alike, but neither registers the other. (b) Correspondence: a is paired with r, and b with r'. The pairing is drawn without direction, since it can be read either way. Neither panel establishes (C), and (C) requires neither.

The direction of C defines two roles. In an instance of C, the contrast depended upon is the *registered* contrast and the contrast that depends is the *registering* contrast. The field that bears the first is the *target* in that instance, and the field that bears the second is the *receiver*. Roles belong to contrasts first, and to fields through them: one field may bear both contrasts, and is then target and receiver in the same instance. Roles are thus positions in a relation, not natures of what occupies them. The statement of (C) does not use the notion of a receiver: C is introduced as a directed relation between contrasts, and the roles are read off it.

Two limitations of (C) should be noted. First, the dependence is not derived from the existence of contrasts; a relation cannot be obtained merely by describing what it relates. It is what the principle of receptivity requires. Second, the dependence is primitive in the core. It is bound by the five conditions stated above, and it is not analysed further. What it consists in within a particular domain is a question for that domain: in physical domains it may be causal, in others it may be logical or constitutive. The core is in this sense a schema. It states what any dependence must satisfy to count as registration, and each domain supplies the dependence; it requires that there be such a dependence, not that it be of any one kind.

6. The Receiving Field

(C) relates two contrasts, and they are two: since no contrast holds in virtue of itself, K' is distinct from K. The second consequence follows.

(R) THE REGISTERING CONTRAST AND ITS BEARER

In an instance of C, the contrast that depends is distinct from the contrast depended upon. What bears it is, insofar as it does, a field; R designates it, and it may be the field that bears the other contrast.

(R) is not a further commitment. It follows from (C), from the condition that nothing holds in virtue of itself, and from what a field is.[6] R is a field in exactly the sense in which F is one: whatever bears a contrast, insofar as it does. Bearing a contrast is not a property peculiar to receivers. What holds of the target therefore holds of R: the contrast concerned is attributable to the field that bears it, and it is the account that picks it out. R itself distinguishes nothing. In particular, it does not distinguish registration from its absence: that K' holds partly in virtue of K is a fact about the two contrasts, not something that R notes.

That the registering is a contrast, and not a single condition, belongs to the second clause of the premise, by the principle of determinacy applied to the registering (Section 2). What follows is that this contrast is a second one, and that what bears it is a field.

R is not a person, a conscious subject, or an independently bounded system. Nothing follows about whether R and F are distinct fields, parts of a larger one, or a single field bearing two contrasts, since a distinction of contrasts does not entail a distinction of substances. A field may register one of its own contrasts, but only through a second contrast.

The positions of K' will be called *conditions* of R and written $r, r', \ldots$. As with the terms of a realized articulation, they are both actual: r and r' are not alternatives between which R is poised, but distinct conditions that are both actual. Nothing requires that K' have the same form as the articulation of F, and no condition is assigned to any particular term of it.

Neither contrast is secured by the definition of the other. Nothing in what a realized articulation is implies that some contrast holds in virtue of its contrast, and a field may bear contrasts that hold in virtue of nothing in F. How widely contrasts are in fact registered depends on what dependences there are; Section 8 shows that on a liberal view of them it is very widely. A field's merely differing within itself is not a registration of anything in F.

(R) also fixes how much individuation the account admits. There must be enough to distinguish K from K', and to take r and r' as positions of one contrast. There need not be enough to identify enduring objects or successive states of a continuing subject: r and r' are both actual positions of one contrast, neither alternatives nor stages in a history. To treat these stronger forms of individuation as already established would build into the foundation a persisting, bounded bearer—which is precisely what an account of a knowing subject must eventually explain.

6. The vocabulary of reception recalls Kant's characterization of sensibility as receptivity, the capacity to acquire representations through being affected by objects (Kant 1998, A19/B33, cf. A50/B74). The receiver introduced here is thinner: it involves no representations, and nothing is claimed about forms of intuition.

7. Registration

Registration can now be defined in a restricted, structural sense. From here on the word is used only as defined. By the premise, every basic epistemic relation is a registration in this sense; not every registration is an epistemic relation.

DEFINITION OF REGISTRATION

A contrast K borne by F is registered by a contrast K' when C holds between them: K' holds at least partly in virtue of K. The field R that bears K' is then said to register K.

Registration is thus nothing over and above an instance of C, and it is always registration *of* a particular contrast: to say that R registers F without qualification is to leave the claim incomplete. An account that states a registration must therefore specify an articulation; the specification is the account's, and the registration does not depend on it. It is also an actual relation between actual contrasts, not a standing disposition of R to respond. Figure 3 depicts the structure so defined.

The definition attributes to the receiver no understanding, identification, or judgement, since none of these capacities has been introduced. Where registration is realized physically, it is found in systems to which nobody would attribute knowledge, and by itself it does not place anything "in the logical space of reasons".[7] Nor does registration move anything from F to R: the receiving contrast need neither resemble the articulation nor reproduce its organization, provided that the dependence holds. Nor, finally, does it require a representational space, or any repertoire of alternatives, in R. Registration relates contrasts that are realized, and nothing in R need stand for conditions that R does not exhibit.

Registration so defined is supplied by neither field in isolation. It comprises three distinct contributions: from the target, the registered contrast; from the receiver, the contrast between its conditions; and between them, the dependence that connects the two. Keeping these contributions distinct prevents registration from being attributed to the target merely because the target is differentiated, or to the receiver merely because its conditions differ. This does not explain the emergence of a knowing subject, but it identifies what such an explanation would have to elaborate.

What is registered, on this account, is always a difference within F, never which way F is. The core provides for registering that a differs from b; it does

7. The phrase is from Sellars (1956, § 36); Brandom (1994, ch. 4) develops the contrast between such reliable differential responsiveness and conceptual capacities. In perception, Dretske's non-epistemic seeing, in which an object is visually differentiated from its surroundings without any belief about it, is a case of registration in the present sense (Dretske 1969).

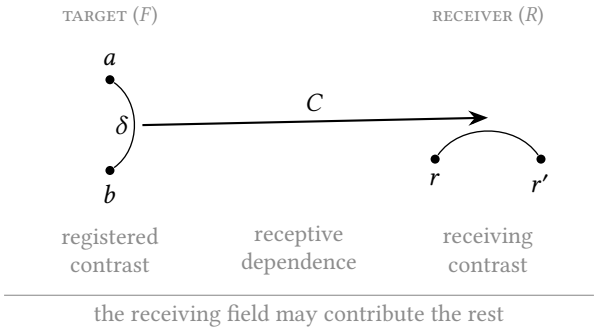


FIGURE 3. Registration. Through C, the contrast between the conditions r and r' of R holds at least partly in virtue of the contrast of the realized articulation $\langle a, \delta, b\rangle_F$. All four terms are actual. The receiving contrast is drawn with a different organization, since it need neither resemble nor reproduce the articulation, and no term is paired with a condition. The arrow runs from contrast to contrast and denotes receptive dependence only: it represents neither a transfer nor an order of occurrences, and the columns are positions in C, not regions.

not provide for registering that F is a rather than b, since that would require b to be an alternative to a rather than another actual term. Nor does it provide for registering that two terms do not differ in a respect: where there is no contrast, there is nothing for (C) to relate, and likeness can enter only through something further, such as two terms answering to one condition. Accounts of information and knowledge built on that richer form presuppose a set of alternatives among which the actual case is singled out.[8] The core admits no such set, since it would require a repertoire of possibilities, held in F, in R, or elsewhere, which is a product of registration and not a precondition of it (Section 2). How registration of which way F is can be built from contrasts between actual terms is left to a later stage. It will need at least a pairing of terms with conditions, which (C) does not include.

8. Two Realizations

The core is a schema, and a schema must be shown to have instances. Two are worked here, chosen so that the dependence is of a different kind in each. They test the schema; they do not fix what it says, and nothing in the preceding sections relies on them.

8. Shannon's theory measures information as selection from a set of possible messages (Shannon 1948), and Dretske's informational content is defined by conditional probabilities over such alternatives (Dretske 1981). In epistemology, relevant-alternatives theories and contrastivism analyse knowledge as the elimination of alternatives (Dretske 1970; Lewis 1996; Schaffer 2004); counterfactual and interventionist analyses of dependence share the same structure (Lewis 1973; Woodward 2003).

A causal case. Let a and b be two bodies that differ in mass, and let them rest on the two pans of a balance, whose ends stand at different heights. K is the contrast between a and b in respect of mass; K' is the contrast between the two ends in respect of height. Both hold, and all four positions are actual. K' holds at least partly in virtue of K, and the dependence is causal. It is partial: K' is due also to how the balance is built. K does not hold, even partly, in virtue of K'. Nothing passes from the bodies to the beam that resembles a difference in mass, and the beam holds no repertoire of the weights that the bodies might have had. The case also shows that registration is indiscriminate: K' holds partly in virtue of the difference in where the two bodies rest on the beam, and so registers that contrast as well. That the dependence here involves a temporal process is a feature of the case, not of the schema; the two contrasts are taken with the balance at rest.

A constitutive case. Let a and b be two actual terms that differ in a fine respect δ, their exact shade, the one being scarlet and the other navy. They then differ also in a coarser respect Δ, their colour, the one being red and the other blue. K is the contrast of a and b in δ; K' is the contrast of a and b in Δ. They are two contrasts, since they have the same positions and different respects. K' holds in virtue of K: the terms differ in colour because they differ in shade. The dependence is not causal, and nothing in it is temporal or spatial; it is the dependence of a difference in a coarser respect on a difference in a finer one. Here K' is due to nothing besides K, so the schema does not require that the receiving field contribute anything. And one field bears both contrasts: it registers one of its own contrasts through a second.

Three results follow. First, the schema is met by dependences of two different kinds. Second, in each case the dependence is identified by the means of its own domain, causal in the one and constitutive in the other. Those means may include the comparison of other cases, or of what would happen otherwise. That is as it should be: such comparisons are evidence of dependence, and the conditions of the core concern what the dependence relates when it holds, not how it is found to hold. Third, the base of the epistemic is not itself epistemic. Wherever a difference in one respect is due to a difference in another, there is a registration, and nobody would call the relation of a difference in colour to a difference in shade an epistemic one. This is not a defect of the schema; it is its main result. There is no primitive epistemic relation. An epistemic relation is an ordinary dependence of difference on difference, of a kind found everywhere, and what makes it epistemic lies in what is built on that dependence: the singling out of one registered contrast, the pairing of terms, order, retention, and whatever further stages require.

9. Registration and Order

The core assumes no order of any kind. Its contrasts are not sequences, and the conditions of R are both actual, not successive. C has a direction, but the direction of a relation between its positions is not an ordering of occurrences: that K' holds in virtue of K does not place either before the other.

The core therefore cannot describe the receiver as *acquiring* a contrast, *retaining* it, or *comparing* it with another. What each of these operations requires is stated here as it appears, for the use of later stages; it is not derived, and each seems to require something different.[9] Comparison requires two receiving contrasts that are both actual; whatever else it requires, it does not require order. Retention requires that one receiver be identified across distinct positions of some ordering. Acquisition requires that a position of such an ordering at which a contrast is not registered stand in an asymmetric relation to one at which it is. Only retention and acquisition require order, and neither follows from (C).

What they require is order in the loosest sense: an asymmetric relation among positions, with no requirement that any two positions be comparable, and with no metric, no linearity, and no universal index. Whether this relation must also be transitive, and so form a strict partial order, may be left open. Temporal succession is one way of realizing such an order; the order of steps in an inference or a proof is another, which is not temporal. The core is compatible with either and requires neither.

For the same reason, if receptive dependence is written with a directed symbol, such as an arrow from F to R, that symbol should not also serve, without further explanation, to represent order. The direction of dependence and the asymmetry of an ordering play different roles in the account and should be kept notationally distinct.

10. Conclusion

Table 1 lists everything the account assumes, by kind, with what each item leaves unestablished.

The reconstruction rests on two principles and on two notions that it does not analyse. Its result is that the basic form of an epistemic relation, in whatever domain it is realized, is a *receptive dependence* of one contrast between actual

9. The three operations closely parallel the synthesis of apprehension, reproduction, and recognition in the first edition of Kant's Transcendental Deduction (Kant 1998, A98–A110), and Husserl's distinction between retention and recollection (Husserl 1991) analyses the second of them. Both treat the operations as presupposing temporal form; the present account requires only order, of which temporal order is one kind.

TABLE 1. What the account assumes, by kind, and what each item leaves unestablished.

ITEM	KIND	WHAT IT LEAVES UNESTABLISHED
Determinacy: to be some way in particular is to differ	Principle, asserted; what "in particular" is taken to mean	Whether the other term is actual; that is settled by receptivity.
Receptivity: the registering is due to what it registers	Principle; marks out the relation analysed	That any such relation occurs.
(P1); (P2), stated exactly as (C); actual terms only	Follow from the two principles	A single registered contrast; a pairing of terms; understanding, representation, order, acquisition, or retention.
That which-way, the lone term's character, and likeness are not basic	Claimed; defence belongs to later stages	How they are built.
Bearing a contrast in a respect	Used without analysis, bound by four conditions	Which contrasts a field bears; what respects there are; any capacity or possibility of the field.
Holding in virtue of	Used without analysis, bound by five conditions	What the dependence consists in within a domain; whether it is transitive; whether two contrasts can depend on each other.
Scope of the reconstruction: specification selects and does not create	Condition of method	That no difference is supplied by a description.
When two contrasts are the same; that nothing holds in virtue of itself	Stipulated	How fields are individuated.
(D); the lone term; (R)	Consequences	That any field bears a contrast apart from registration; that R and F are distinct in substance.
The logic of the exposition and its modal words; what is actual; that what a contrast is due to can be several things; the account's own reference to its targets	Presupposed	Not examined here.

terms on another, with target and receiver distinguished as positions within that dependence. The conditions are necessary with the necessity of an analysis: they hold of every receptive relation to something in particular. They are not sufficient for perception or for knowledge, and nothing is claimed here about aboutness by description.

The core is minimal, but it is not empty. One thing it rules out, and several things it shows to be no part of what the premise requires. It rules out that registration consists in resemblance or in a correspondence between terms, since neither supplies the dependence that the second clause of the premise requires (Section 5). It leaves outside the reconstruction, as a matter of scope, any respect whose holding depends, even partly, on the specification or its notation (Section 4); and a field's merely differing within itself, where its contrast does not hold even partly in virtue of the target's contrast, is not a registration of anything in the target. For the rest, the core does not include, among the conditions of registration, a transfer from target to receiver, since the receiving contrast need neither resemble nor reproduce the articulation (Section 7); a repertoire of alternatives, which is a product of registration (Section 2); time, or order of any kind, neither of which appears in the definition (Section 9); or a knowing subject, concepts, or representations, none of which figures in the premise or in what follows from it. That these are not among the conditions does not show that registration can occur without them in a given domain, and none of this denies that such structures are present in particular epistemic relations. What is claimed is only that the premise does not require them: wherever they are needed, they must enter as further commitments, stated as such. Nor does the core exclude that a registered contrast depends in turn on the contrast that registers it; it describes that as registration in both directions (Section 5).

The account can be refused in principle at two points: at determinacy and at receptivity (Section 2). A reader who accepts both but rejects the three arguments about alternatives keeps the two conditions, a difference and a dependence on it, and changes only their form. What the account leaves undone should be stated as plainly. First, the two notions on which it rests are bound by stated conditions but not analysed. The bearing of a contrast carries the whole weight of the distinction between a contrast of the target and one supplied by a description, and the account does not say what respects there are. Second, the core does not provide for a relation to which way something is as against an alternative, to anything of a term that differs from nothing actual in the respect concerned, or to the fact that two terms do not differ; it claims that these are built on registration and does not yet show how. Third, registration as defined is indiscriminate: a contrast registers every contrast it holds at least partly in virtue of, and since transitivity is left open, the core does not say whether a registering reaches beyond what it immediately depends on. Fourth, the core says what receptive dependence is, not how an instance of it is known to be one, and it sets no bound on what a receiving field contributes; both belong to

later stages. Fifth, "actual" is not tensed. Whether what is earlier and what is later are both actual, and so whether there can be a contrast between them, is not settled here and belongs with the question of order. Sixth, the account picks out the contrasts of its targets by description, and the guiding premise does not cover aboutness by description: the account uses a relation to its targets that it does not itself cover.

Five questions remain open. The first is what the dependence C consists in, and what gives it its direction, in a given domain. The second is whether receptive dependence can itself be explained in terms of more elementary relations, whether it is transitive, and whether two contrasts can depend on each other; an arrow whose meaning is left unspecified would not answer it. The third is how registration of which way F is, of the character of a lone term, and of likeness are built from contrasts between actual terms. The fourth is what order acquisition and retention require, whether anything stronger than an asymmetric relation is needed, and whether the dependence itself, if it is transitive, supplies that order. The fifth is whether aboutness by description, of the future, and of mathematical objects is built from registered contrasts. The reconstruction ends at structural registration. Order, memory, recognition, and representation have not been presupposed in reaching it, and none of them has yet been derived from it.

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